ChE 2O04 Tutorial #2

Tutorial Problem #1. In the table below are shear stress and shear rate data for three bottles of syrup identified as A, B, & C (Note that the data are also posted on Avenue in a text document). One bottle is supposed to contain pure grade 'Canada #1 very clear' maple syrup (which as we discussed in lecture is a Newtonian fluid) while the other two are thought to be cheaper grade 'pancake syrups' (i.e. corn syrup plus colouring and other additives) that have some non-Newtonian rheological behaviour.

a. Analyze the data using the linear regression function in Excel to determine which bottle (if any) contains the 'pure' maple syrup; be sure to give a clear description of the exact reasons how you conclude that the syrup is a Newtonian or non-Newtonian fluid.

b. For the pure maple syrup, calculate the viscosity (in cP units).

Bottle A		Bottle B		Bottle C	
Shear Rate (1/s)	Shear Stress (Pa)	Shear Rate (1/s)	Shear Stress (Pa)	Shear Rate (1/s)	Shear Stress (Pa)
80	45	80	56	80	51
120	71	120	77	120	72
160	98	160	101	160	99
200	124	200	126	200	121
240	148	240	152	240	148
280	173	280	175	280	171
320	195	320	200	320	197
360	216	360	226	360	218
400	238	400	250	400	246

Tutorial Problem #2. Using the Colebrook Equation (given below), create a nice figure in Excel that shows how the friction factor (at relative roughness = 0.004) changes as the Reynolds number increases from 8,000 to 8,000,000 .

Hint: The quality of your figure wll be greatly improved if you use a logarithmic scale for the axis with the Reynolds numbers.
Hint: It is generally a good idea that for a logarithmic scale to include the minor axis gridlines. Can you rationalize why this is?

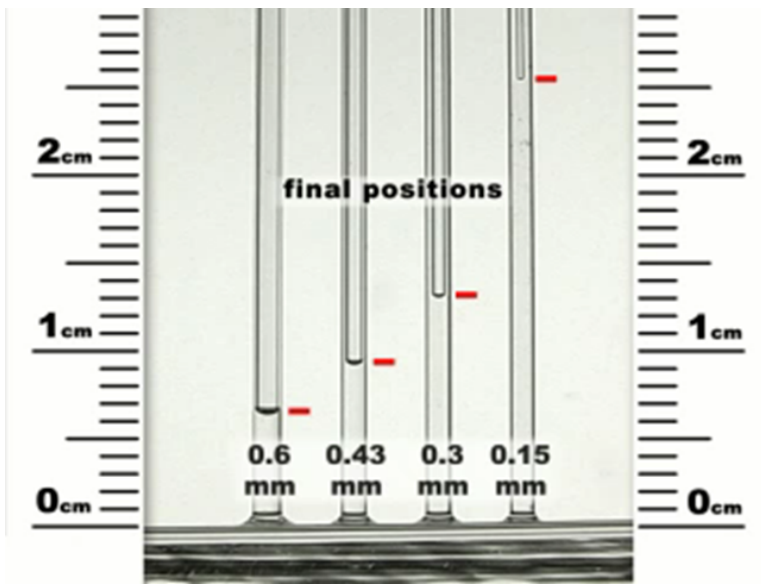
$$\frac{1}{\sqrt{f}} + 2.0\log_{10}\left[\frac{\varepsilon}{3.7} + \frac{2.51}{Re\sqrt{f}}\right] = 0$$

Tutorial Problem #3.

a. Create a nice figure in Excel that shows the profile for height of water (at 20 oC) inside a capillary tube varies with the radius of the capillary tube (for range 0.1 to 5 mm) as determined by:

$$\Delta h = \frac{2\sigma \cos\theta}{\rho g r_{capillary}}$$

when $\theta = 75°$
$\rho = 998\ \mathrm{kg/m^3}$

HINT: You can find surface tension values for water in Table A.8

b. Make a new figure that has the profile from part a) along with the four experimental values shown in the image to the right. Then provide a short commentary on the overall agreement between the predicted values (from equation in part a) and the experimental values.